Lakshman Raj Baskaran | Software Engineer

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Summary

Software Engineer with an M.Sc. in Computer Science from Trinity College Dublin and an Electronics and Communication Engineering background, building **agentic AI systems**, **LLM evaluation pipelines** and the **Python and Java backends** behind them (**FastAPI**, **Spring Boot**, **PostgreSQL**, **Docker**, **Azure**), alongside **statistical modelling in R** and **embedded IoT** work on **ESP32**. Microsoft Certified Azure AI Cloud Developer Associate and Anthropic MCP certified.

Experience

AI Workflow Engineer Intern, FOREO Sep 2026 - Nov 2026

Python, Qwen via Ollama, Pydantic, HDBSCAN, scikit-learn, SQLite/SQLAlchemy, pytest, GitHub Actions

- Building the **FAQ Gap Analyser**, a tool that checks how well FOREO's published FAQs answer real customer questions, using **2,584** Turkish and English marketplace questions across **93** products.
- Found that the FAQs answer only about **40%** of these questions; **8 of the 10** biggest gaps the tool flagged were confirmed by manual review.
- Grouped similar questions into **16 topics** with multilingual **embeddings** and **HDBSCAN**, and used an **LLM** to judge whether an FAQ actually answers each one; its verdicts matched human labels **86.7%** of the time on held-out data.
- Designed evaluation on **synthetic and real hold-out splits**, with a **provenance rule** that keeps synthetic data out of any business result.

Engineering Intern, Siemens Centre of Excellence Aug 2023 - Sept 2023

Industrial Automation Systems, India

- Delivered a **warehouse management automation** project in **PLC ladder logic** inside a small engineering team, coordinating on shared control logic so that separately written pieces behaved correctly once combined.
- Documented the module and presented it to engineers, in an industrial setting where a mistake in control logic has physical consequences.

Technical Skills

Languages: Python, Java, SQL, C++, R, Arduino C++, PL/SQL

AI and Agents: Anthropic Claude API, Deep Agents, LangChain, Model Context Protocol (MCP), Spring AI, Ollama (Qwen), LangSmith (tracing, evaluation), multi-agent systems, tool design, human-in-the-loop workflows, LLM-as-judge evaluation, Claude Code, Codex

Machine Learning and Data: PyTorch, Graph Neural Networks, scikit-learn, embeddings (e5-large-v2), clustering (UMAP, HDBSCAN), dense retrieval, statistical modelling (GAMs, R sf), Google BigQuery, dbt (basic), Excel (formulas, pivot tables, lookups), Power BI (basic)

Backend: Spring Boot (REST, JPA, AOP, WebClient), FastAPI, Flask, Celery, Pydantic, SQLAlchemy, OpenAPI/Swagger, Playwright (scraping), Chainlit

Databases: PostgreSQL, SQLite, Redis, Azure Cosmos DB, Cloudflare D1

Cloud and DevOps: Docker, GitHub Actions (CI/CD), Azure Container Apps, Azure Cosmos DB, Render, Cloudflare Pages and R2, Kubernetes (minikube), Terraform (basic), Azure DevOps (basic), Prometheus, Grafana, Linux, Git, Maven

Azure (basic): AKS, ACR, Azure Managed Redis, Azure Service Bus, Key Vault, Entra ID, Azure Monitor, PowerShell

Testing: pytest, JUnit 5, AssertJ, FastAPI TestClient, unit and integration testing

Embedded and Hardware: ESP32, ESP-EYE, LoRa, Raspberry Pi, sensor integration, PLC ladder logic, VLSI, COMSOL

Tools: Git, IntelliJ IDEA, Confluence

Projects

AI Agent System (Revenue Manager) | *Python, Deep Agents, Anthropic Claude API, PostgreSQL, Playwright, Chainlit*

- Built an **agentic AI revenue-manager** for hotels answering revenue queries in real time, using **7 tools, 2 subagents, and 11 domain skill files**, on the **Deep Agents framework** with **Claude Sonnet API**, and **human-in-the-loop** approval on sensitive queries.
- Cut latency **58% (57.5s → 24.1s)** by using **LangSmith tracing** to root-cause **context accumulation across shared threads** (24k tokens growing to 226k) and moving to per-session threads.
- Reduced **LLM calls** on one-tool questions by replacing vague prompts with a **deterministic routing table**.
- Built a full **data pipeline and UI**: **Playwright** web scraper, **ETL into PostgreSQL**, **Chainlit** front end, with **50 unit tests** across tools, pipeline, skills and agent wiring, including checks of tool output against direct SQL.

- Deployed it on **Render** behind password login with a **health endpoint** that fingerprints the live data, and fixed a **race condition** with a per-session lock and a human-in-the-loop double-call bug.

Spring Introspection MCP Server | *Java 21*, *Spring Boot 4*, *Spring AI 2*, *MCP*

- Built a **Spring Boot starter** for AI coding agents exposing **7 MCP tools**, resolving **property precedence chains** and **auto-configuration condition outcomes** that depend on **runtime env vars and classpath**, invisible in static source.
- Cut token usage **88% (80.7K → 9.5K tokens/query)** on **configuration and bean-provenance** questions by replacing multi-step **Actuator/source exploration** with **single tool calls**, benchmarked against a **naive source-reading agent**.
- Built **proxy and advice introspection** revealing a bean's real **runtime proxy type** and **applied advisors**, surfacing issues like the **Spring self-invocation problem**, validated with **79 end-to-end tests** against a live application.
- Designed a **deny-over-allow redaction** layer so passwords, tokens and other secrets never reach the agent, covered by a **17-case** test suite.

Disaster Management and Citizen Safety Platform | *Java Spring Boot*, *Python FastAPI*, *PostgreSQL*, *Next.js*, *Docker* Sep 2024 - Mar 2025

- Built a **citizen safety platform** in a team of 8 which acted as a single source of truth during disasters, ingesting **incident reports**, dispatching the right **emergency services**, and **rerouting people and traffic** away from affected zones.
- Owned part of the **Java Spring Boot backend** coordinating emergency response, integrating **TomTom and OpenStreetMap** for incident location and rerouting data, and contributed to the **Next.js admin dashboard**.
- Eliminated a **single point of failure** in resource allocation with a **deterministic fallback**, so every report still gets a dispatchable **fire, medical, and police allocation** when the upstream AI fails.
- Replaced one blended severity score with **3 per-service scoring models**, so a crowded but injury-free incident escalates police without dispatching unneeded ambulances.
- Set up **CI/CD on GitHub Actions** building **Docker images** for the platform's services.

Depression and Voting Patterns: Spatial Statistical Model | *R*, *sf*, *GAMs*, *Trinity College Dublin* 2025

- Tested whether county depression rates predict Republican vote share across **3,073 US counties** in **50 states**, joining election, health and census data and matching every county to its map position using **Census shapefiles** in **R (sf)**.
- A simple model suggested each 1-point rise in depression lifted the odds of a Republican vote by **~17%**; built **6 progressively stronger spatial models** controlling for location, state and race, and showed the real effect was **~2%**, though still significant.

Wildfire Detection and Suppression System | *ESP32*, *Arduino C++*, *LoRa*, *Cloud ML* Jan 2025 - Mar 2025

- Built an end-to-end **IoT** system: an **ESP32** sensor unit reads **MQ-2 gas**, **DHT11 temperature/humidity**, **ultrasonic** and temperature sensors, and an **ESP-EYE** camera captures images of the area.
- Relayed sensor data and images over a **LoRa** network, node to node, to a gateway at the forest edge that uploads to the **cloud**, where an ML model detects fire.
- Built an actuator unit that receives the cloud result, uses a **magnetometer** to orient towards the sensor unit and a **flame sensor** to aim a water spray.

Education

Trinity College Dublin Sep 2024 - Sep 2025

M.Sc. Computer Science, Future Networked Systems, Dublin, Ireland (NFQ Level 9)

Thesis: Automatic Modulation Classification Using Graph Neural Networks (supervisor: Dr. Viet Quoc Pham)

Modules: Artificial Intelligence, Machine Learning, Advanced Software Engineering, Internet of Things, Applied Statistical Modelling, Scalable Computing, Distributed Systems, Urban Computing, Next Generation Networks, Security and Privacy, Research and Innovation Methods

SRM Institute of Science and Technology 2020 - 2024

B.Tech. Electronics and Communication Engineering, 8.52/10, India

Research on photonic biosensors for early detection of basal cell carcinoma (skin cancer) cells, as a real-time alternative to traditional detection.

Modules: VLSI, Embedded Systems, Microprocessors and Microcontrollers, Cryptography

Indian Institute of Technology Madras Jan 2021 - Jan 2026

Diploma in Programming (online), India

Modules: Object-Oriented Programming in Java, Python, Data Structures and Algorithms, Modern Application Development I and II, System Commands

CBSE Class 12

418/500, India

Certifications and Publications

- Microsoft Certified: Azure AI Cloud Developer Associate, Microsoft, Aug 2026 (Credential ID: 6FC7792BA214B975).
- Introduction to Model Context Protocol (MCP), Anthropic, Apr 2026 (Credential ID: 9mfdk4ay7dwj).
- Co-authored a peer-reviewed book chapter, Advanced Materials for Biomedical Devices (2024), on photonic biosensor design using COMSOL and regression analysis.

Activities and Languages

- Took part in **Model United Nations** and helped organise MUN conferences.
- Helped organise a volunteer group for a **beach clean-up**.
- Plays **hockey**.
- English: IELTS 8.0 overall.